



In the 1940s, aircraft were used to spray DDT over the widest possible areas. Its polluting effects lasted for many years.

2 MILLION TONS THE AMOUNT OF DDT USED THROUGHOUT THE WORLD SINCE 1940

of the war effort. He used a material called polystyrene—a type of plastic originally made from tree resin—and chemically treated it to produce countless bubbles in the manufacturing process. The resulting foam polystyrene, trademarked as **styrofoam**, was inexpensive and lightweight.

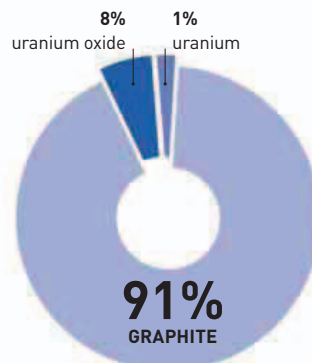


THE ROLE OF ATP IN METABOLISM

Plant and animal cells contain powerhouses called mitochondria (sepia). They are packed with membrane folds that carry the molecular machinery needed to make a chemical called adenosine triphosphate (ATP) from high-energy foods. Energy from ATP is released to order to drive cellular activity, such as building DNA and protein.

IN 1939, SWISS CHEMIST PAUL HERMANN MÜLLER (1899–1965) had discovered that a chlorine-containing chemical called **DDT** (dichlorodiphenyltrichloroethane) was lethal when it came in contact with insects, and could possibly be used to control insect pests. In September, the US received the **first stocks of DDT to begin using it on a wide scale**. During World War II, DDT was used widely to control lice-born typhus as well as malaria-carrying mosquitos. In the post-war years, DDT application increased further as agriculturalists began using it to kill crop-eating pests. However, by the 1960s, the world had realized that DDT poisoned the environment by accumulating in food chains [see 1962], and the chemical that had earlier earned Müller a Noble Prize was **banned from most countries**.

Two American biologists, German-born **Max Delbrück** (1906–81) and Italian-born **Salvador Luria** (1912–91), embarked on a collaborative study of **bacteriophages** (also known as phages)—viruses that infect and



Enrico Fermi's nuclear reactor
Neutron-releasing uranium pellets were at the heart of the nuclear chain reaction in Fermi's reactor. Graphite blocks slowed the neutrons.

kill bacteria. They investigated the process whereby some bacteria mutate to become genetically resistant to infection, and in 1943, explained that this happened spontaneously, and was not induced by the environment.

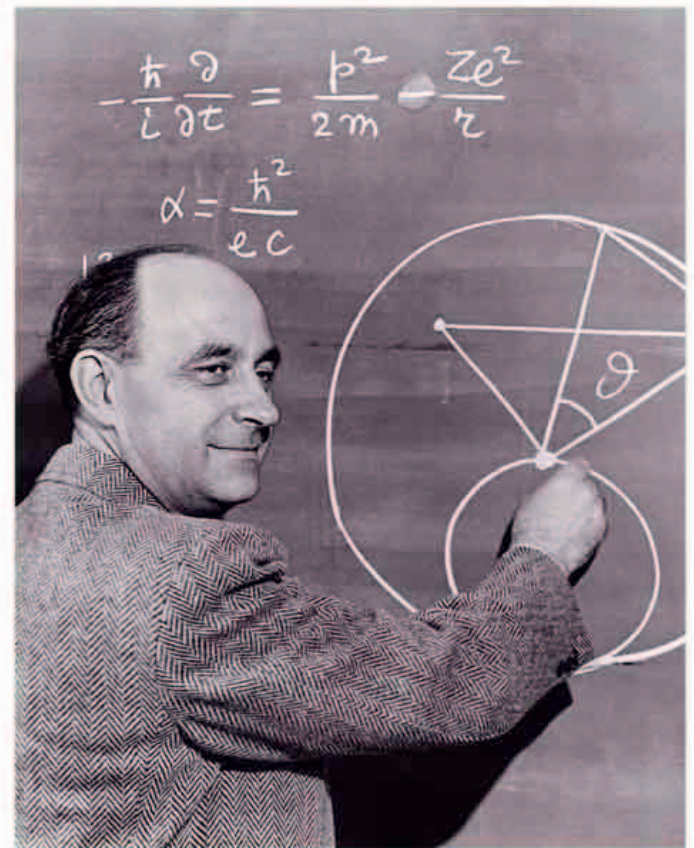
Scientists had already demonstrated the practical possibility of a self-sustained nuclear reaction in a controlled setting [see 1938]. In December, Italian-American physicist **Enrico Fermi** oversaw the operation of the world's **first nuclear reactor**—

Chicago pile-1—built beneath a football stand at the University of Chicago. The reactor bombarded a sample of uranium with neutrons to **trigger a chain reaction**, which was controlled with neutron-absorbing cadmium rods. It ran successfully for four and a half minutes before Fermi stopped the process. This event marked a critical step in the

Manhattan Project—the US government-led Allied research and development initiative that produced the first atomic bombs during World War II.

Enrico Fermi

In 1938, Enrico Fermi moved to the US with his Jewish wife to escape Italy's anti-Semitic policies. He received the Nobel Prize for Physics the same year.



“ THE ITALIAN NAVIGATOR HAS LANDED IN THE NEW WORLD. ”

Arthur Compton, Director of the Metallurgical Laboratory, University of Chicago, in a coded message referring to Enrico Fermi's success, 1942

November 15, 1941 American scientists George Beadle and Edward Tatum show that genes produce their effects by regulating particular **enzymes**

February 27 English physicist James Stanley Hey detects **radio waves** from the Sun

September USA receives first samples of **DDT** for wide-scale use, to control typhus from lice

October 3 Swedish physicist Hannes Alfvén describes **electromagnetic waves** in liquids

October 3 German engineers Wernher von Braun and Walter Dornberger perform the first successful test launch of a **V2 rocket**

December 2 Enrico Fermi oversees the **first self-sustained nuclear chain reaction** in the first nuclear reactor

Max Delbrück and Salvador Luria demonstrate that **genetic mutations** arise spontaneously in bacteria